Basin2Vec: Learning Basin-Centric Embeddings for Scalable Hydrological Modeling

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Abstract

Reliable streamflow prediction across time and in ungauged basins requires representations that capture hydrologically relevant basin characteristics. Existing methods rely on either fixed handcrafted attributes or general-purpose Earth observation embeddings not tailored to basin structure. We introduce ***Basin2Vec****, a self-supervised framework that learns compact basin representations from hydroclimatic and physiographic data without observed streamflow. Basin2Vec organizes representation learning around basin scale and relationships across space and time. Across multiple forecasting architectures, it consistently outperforms predefined attributes and general-purpose geospatial embeddings in temporal generalization and zero-shot spatial transfer. Basin2Vec provides average relative gains of up to 8.3% temporally and achieves 3.5× the median predictive performance of the strongest baselines on unseen basins. These results demonstrate the effectiveness of basin-centric pretraining for transferable hydrologic prediction. Code is available at* https://github.com/Muhammad-Talha-MT/basin2vec.

1. Introduction

Streamflow, or river discharge, is the volume of water passing through a river cross-section per unit time. Accurate streamflow prediction is critical for flood forecasting, drought monitoring, and water resources planning. However, streamflow observations are available at only a limited and unevenly distributed network of gauging stations [6, 30, 35]. Therefore, hydrological models must generalize in two ways: *temporal generalization*, in which they predict future streamflow in basins seen during training, and *spatial generalization*, in which they predict streamflow in basins that were not used for training [10, 18]. Accurate prediction in both settings requires a robust basin representation.

Table 1. Comparison of basin representations across three design requirements. Scores indicate High, Moderate, or Low support.

Approach	Basin-indexed	Spatial & scale-aware	Hydrologic relation-aware
CAMELS Attr. [1]	●●●	●●●	●●●
AlphaEarth [5]	●●●	●●●	●●●
TESSERA [11]	●●●	●●●	●●●
SatCLIP [16]	●●●	●●●	●●●
Basin2Vec (Ours)	●●●	●●●	●●●

●●● High ●●● Moderate ●●● Low

In particular, the representation must capture the *dynamic meteorological inputs*, such as precipitation and temperature, along with a *static basin descriptor* that distinguishes characteristics of the target basin from others. Such a descriptor must summarize the factors governing how a basin converts weather into streamflow, including climate, terrain, soils, land cover, subsurface properties, and drainage area [1, 8, 24, 34]. An informative basin representation is particularly important for spatial generalization because it enables the model to identify similarities and differences between observed and previously unseen basins.

Regional streamflow models commonly represent each basin as a fixed-length feature vector derived from datasets such as CAMELS and GAGES-II [1, 8, 9, 19]. While these compact attributes may summarize the spatial characteristics of the basin using statistics such as mean elevation or forest fraction, they do not capture the spatial variation of these characteristics within the basin, causing the downstream models to be oblivious to their spatial heterogeneity. Gridded datasets such as CAMELS-SPAT preserve this information but introduce high-dimensional spatial inputs [17]. This motivates learning compact representations directly from basin-aligned geospatial data.

General-purpose Earth observation (EO) embeddings provide reusable representations of satellite observations and geographic locations [5, 11, 16]. However, they are pre-

trained on pixels, image patches, or coordinates rather than drainage basins, requiring post hoc aggregation to construct basin descriptors. Moreover, their pretraining objectives do not explicitly encode physical basin scale or hydrologically meaningful relationships among basins.

Together, these limitations motivate three design requirements for basin-centric representation, as summarized in Table 1. First, the representation should be *basin-indexed*, meaning that its entities correspond to drainage basins rather than individual pixels or coordinates. Second, the learned embeddings must be *spatial- and scale-aware*, meaning that it learns from within-basin spatial inputs while retaining information about physical basin scale. Finally, it should be *hydrologic relation-aware*, meaning that the learned embedding geometry reflects the hydrological and physiographic relations among the basins.

To meet these requirements, we introduce **Basin2Vec**. Our contributions are threefold. **(1)** We formulate basin-centric self-supervision around a *basin-year* entity, producing reusable descriptors from gridded hydroclimatic and physiographic data without observed streamflow. **(2)** We propose a *scale-aware encoder* that preserves spatial and physical-scale information, together with a *relation-aware contrastive objective* that models temporal, overlapping, and nested basin relationships. **(3)** We evaluate the resulting descriptors with five forecasting architectures under temporal generalization and zero-shot spatial transfer. Basin2Vec consistently outperforms predefined attributes and general-purpose EO embeddings, with average relative temporal gains of up to 8.3% and 3.5× the median predictive performance of the strongest alternative on unseen basins.

2. Related Work

Basin characterization in regional hydrological models. Large-sample hydrological models are commonly conditioned on curated basin descriptors from CAMELS or GAGES-II [1, 8, 9, 26]. This paradigm, including entity-aware and regional LSTMs, has demonstrated the value of static attributes for shared prediction across multiple basins [10, 18, 19]. CAMELS-SPAT extends this foundation with basin-aligned rasters of climatic and physiographic variables [17], but using them directly requires each downstream model to process high-dimensional spatial inputs.

Earth-observation representation learning. General-purpose EO representation-learning methods derive transferable features from remotely sensed or geographic data. SatMAE learns visual representations from satellite image patches [7]; AlphaEarth and TESSERA learn geospatial representations from EO observations; and SatCLIP learns location representations from geographic coordinates [5, 11, 16]. Recent studies have used these representations as inputs to hydrological models [21, 27, 28]. Because these representations are natively associated with pixels, patches, or coordinates, their application to hydrological basins typically requires post hoc spatial aggregation.

Hydrology-specific representation learning. Previous hydrology-specific learning methods have used encoder-decoder or autoencoder to infer hydrological signatures and landscape features from meteorological and streamflow time series [2, 4]. KG-SSL similarly learns basin representations from basin-level meteorological inputs and observed streamflow, treating samples from the same basin across different years as related and samples from different basins as unrelated [13]. However, these methods rely on observed streamflow during the representation learning, limiting their applicability to ungauged basins.

3. Basin2Vec

3.1. Overview

Basin2Vec learns basin-level representations from basin-year entities. Each entity corresponds to a basin-year tuple, (b, y), along with its monthly hydroclimatic rasters $X^{\mathrm{dyn}}_{b,y}$, static or annual spatial variables $X^{\mathrm{sta}}_{b,y}$, binary basin mask M_b, and basin metadata X^{meta}_{b}. Note that each basin has been resampled to a fixed $H \times W$ patch using the approach described in [33], although its boundary is irregular and its physical area may vary substantially. Details of the resampling approach are given in Appendix A.2. The monthly input has shape $X^{\mathrm{dyn}}_{b,y} \in \mathbb{R}^{12\times C_d\times H\times W}$, where C_d is the number of hydroclimatic variables. The quasi-static input has shape $X^{\mathrm{sta}}_{b,y} \in \mathbb{R}^{C_s\times H\times W}$, where C_s is the number of static or annually varying physiographic variables. As shown in Figure 1, Basin2Vec uses a **scale-aware encoder** to map these masked basin-aligned inputs to a normalized basin-year embedding $z_{b,y} \in \mathbb{R}^d$, and a **relation-aware contrastive objective** to structure the embedding space using same-basin, temporal, spatial overlap, nesting, and physiographic relations. After pretraining, the basin-year embeddings $z_{b,y} : y \in \mathcal{Y}_b$ are averaged across the available years $\mathcal{Y}_b$ to obtain the fixed basin-level descriptor $\bar{z}_b \in \mathbb{R}^d$ for downstream streamflow prediction.

3.2. Scale-Aware Basin Encoder

The encoder generates the basin-year embedding $z_{b,y}$ by processing the inputs defined in Section 3.1 according to the following steps: mask-aware spatial encoding of the monthly and quasi-static inputs, fusion of the two resulting representations, and incorporation of basin metadata. Since the basins have been resampled to the same spatial dimensions, which removes the information about its physical size, an auxiliary drainage-area objective is also applied

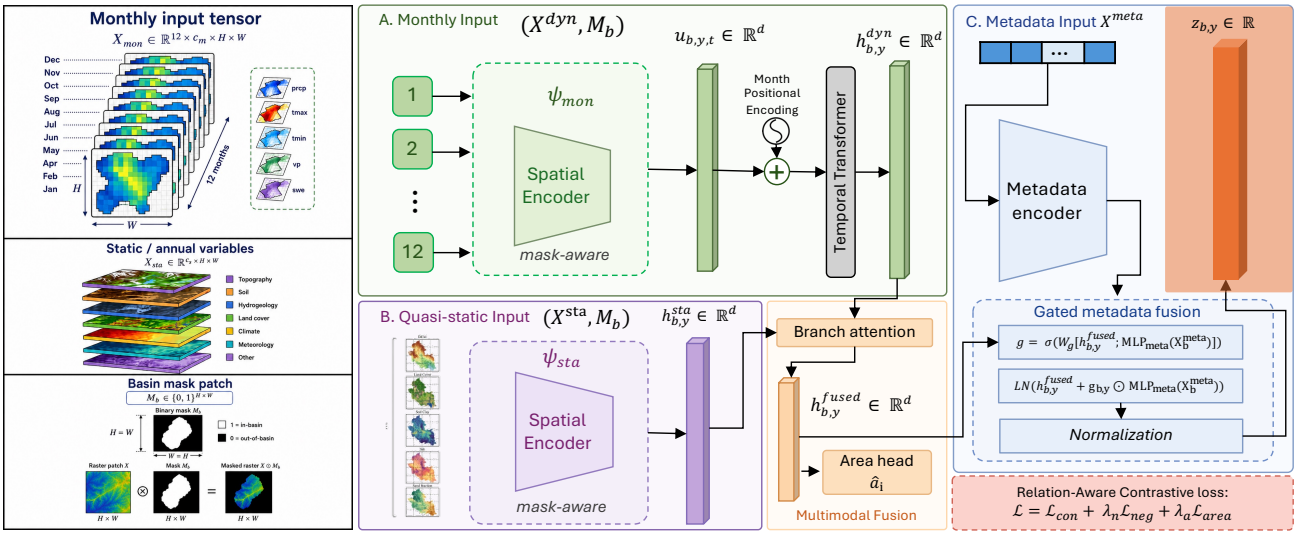


Figure 1. **Overview of Basin2Vec.** The scale-aware encoder maps monthly and quasi-static basin-aligned grids, masks, and metadata to normalized basin-year embeddings $z_{b,y}$ through mask-aware encoding, multimodal fusion, and area regularization. during pretraining.

Mask-Aware Spatial Encoding. Each basin-aligned input contains grid cells both inside and outside the irregular basin boundary. To prevent out-of-basin cells from affecting the representation, Basin2Vec applies the binary mask M_b during spatial encoding and pooling. Given a multi-channel input tensor X, whose last two dimensions correspond to the $H \times W$ spatial grid, a convolutional neural network (CNN) is used to extract a spatial feature map F from the masked input. The feature map is then aggregated via masked average pooling $\mathrm{AvgPool}_{M_b}$ and mask-normalized attention pooling $\mathrm{AttnPool}_{M_b}$, as shown below:

$$\begin{aligned} u &= \psi(X, M_b) \\ &\equiv \mathrm{MLP}\left(\left[\mathrm{AvgPool}_{M_b}(F); \mathrm{AttnPool}_{M_b}(F)\right]\right), \end{aligned} \tag{1}$$

where ψ denotes the mask-aware spatial encoder, $F = \mathrm{CNN}(X \odot M_b)$ denotes the masked spatial feature map, and $\odot$ denotes element-wise multiplication. $\mathrm{AvgPool}_{M_b}$ averages the features across all valid basin cells, whereas $\mathrm{AttnPool}_{M_b}$ assigns learned weights to these cells and normalizes the weights only within the basin boundary. The two pooled summaries are concatenated and transformed by an MLP to produce the spatial basin token $u \in \mathbb{R}^d$.

The mask-aware spatial encoder ψ is applied separately to the dynamic and quasi-static inputs, each with its own learned parameters. For each month $t \in \{1, \ldots, 12\}$, the dynamic encoder ψ_{dyn} maps $X^{\mathrm{dyn}}_{b,y,t}$ to a monthly basin token $u_{b,y,t}$. A learned month embedding $p_t \in \mathbb{R}^d$ is added to identify the token's position in the annual sequence, following standard positional-encoding practice [36]. A Transformer with four attention heads and GELU feed-forward blocks then models its temporal relationships across the 12 months. The contextualized monthly tokens are aggregated to produce the annual dynamic representation:

$$\begin{aligned} u_{b,y,t} &= \psi_{\mathrm{dyn}}\left(X^{\mathrm{dyn}}_{b,y,t}, M_b\right), \\ \{\widetilde{u}_{b,y,t}\}_{t=1}^{12} &= \mathrm{Transformer}\left(\{u_{b,y,t} + p_t\}_{t=1}^{12}\right), \\ h^{\mathrm{dyn}}_{b,y} &= \mathrm{MLP}_{\mathrm{dyn}}\left(\sum_{t=1}^{12} \beta_{b,y,t}\, \widetilde{u}_{b,y,t}\right). \end{aligned} \tag{2}$$

Here, $\widetilde{u}_{b,y,t}$ is the contextualized token for month t. The weight $\beta_{b,y,t}$ is produced by a learned attention-pooling operation and normalized across the 12 months such that $\sum_{t=1}^{12} \beta_{b,y,t} = 1$. The resulting vector $h^{\mathrm{dyn}}_{b,y} \in \mathbb{R}^d$ summarizes the monthly hydroclimatic inputs for the basin-year.

The quasi-static input $X^{\mathrm{sta}}_{b,y}$ is represented by a single multi-channel tensor for each basin and therefore does not require temporal processing. A separate mask-aware spatial encoder ψ_{sta} processes its channels jointly with an MLP to produce the quasi-static embedding:

$$h^{\mathrm{sta}}_{b,y} = \mathrm{MLP}_{\mathrm{sta}}\left(\psi_{\mathrm{sta}}\left(X^{\mathrm{sta}}_{b,y}, M_b\right)\right). \tag{3}$$

Here, $h^{\mathrm{sta}}_{b,y} \in \mathbb{R}^d$ summarizes the static and annually varying physiographic inputs for basin b in year y.

Fusion and Metadata Conditioning. The dynamic and quasi-static representations are fused into a single spatial representation using learned branch-attention weights:

$$\begin{aligned} \alpha^{\ell}_{b,y} &= \mathrm{softmax}_{\ell \in \{\mathrm{dyn,sta}\}}\left(W_\alpha h^{\ell}_{b,y}\right), \\ h^{\mathrm{fused}}_{b,y} &= \sum_{\ell \in \{\mathrm{dyn,sta}\}} \alpha^{\ell}_{b,y} h^{\ell}_{b,y}. \end{aligned} \tag{4}$$

Here, W_α is a learned scoring layer, and the weights $\alpha^{\mathrm{dyn}}_{b,y}$ and $\alpha^{\mathrm{sta}}_{b,y}$ are normalized to sum to one. The resulting vector $h^{\mathrm{fused}}_{b,y} \in \mathbb{R}^d$ combines the spatial information learned from the static and dynamic input modalities.

Since all the basins have been mapped to the same $H \times W$ grid, their rescaled inputs do not directly convey physical

basin size. Basin2Vec therefore uses an auxiliary prediction head q_a to estimate the standardized log drainage area from the fused spatial representation by minimizing the following smoothed L1 loss:

$$\mathcal{L}_{\text{area}} = \frac{1}{B}\sum_{i=1}^{B} \ell\left(\hat{a}_i - a_i\right), \qquad \ell(e) = \begin{cases} \frac{1}{2}e^2, & |e| < 1, \\ |e| - \frac{1}{2}, & |e| \geq 1. \end{cases} \tag{5}$$

Here, B is the mini-batch size, a_i is the standardized log drainage area of the basin associated with sample i, $\hat{a}_i = q_a(h_i^{\text{fused}})$ is its log drainage area estimated by the prediction head q_a, and $e_i = \hat{a}_i - a_i$ is the prediction error. The loss is quadratic for small errors and linear for large errors, reducing sensitivity to extreme differences in basin area. Applying this objective to the fused representation h^{fused} encourages the spatial encoder to retain information about the physical basin scale.

Finally, basin-level metadata X_b^{meta}, including drainage area, basin geometry, and effective patch resolution, are also incorporated into the embedding through gated conditioning. An MLP initially maps the standardized metadata to the embedding dimension, while a dimension-wise gate g controls their contribution to the fused spatial representation:

$$\begin{aligned} g_{b,y} &= \sigma\left(W_g\left[h_{b,y}^{\text{fused}}; MLP_{\text{meta}}(X_b^{\text{meta}})\right]\right), \\ v_{b,y} &= \gamma\left(\text{LN}\left(h_{b,y}^{\text{fused}} + g_{b,y} \odot MLP_{\text{meta}}(X_b^{\text{meta}})\right)\right), \\ z_{b,y} &= \text{Normalize}\left(q_\phi(v_{b,y})\right). \end{aligned} \tag{6}$$

Here, $[\cdot;\cdot]$ denotes concatenation, σ is the sigmoid function, and $g_{b,y} \in [0,1]^d$ is the metadata gate. The gated metadata are added to $h_{b,y}^{\text{fused}}$, after which layer normalization and another MLP γ is applied to produce the metadata-conditioned representation $v_{b,y}$. The projection head q_ϕ maps this representation to the embedding space, and ℓ_2 normalization produces the final basin-year embedding $z_{b,y} \in \mathbb{R}^d$.

3.3. Relation-Aware Contrastive Objective

Standard contrastive learning treats augmented views of the same entity as positive examples and all other entities as negative examples. This assumption is insufficient for basin-year entities that could be related through basin identity, temporal proximity, or spatial relationship. To overcome this limitation, Basin2Vec considers three positive relations between basin-year entities: (1) a *hard positive* between two augmented views of the same basin-year entity, (2) a *temporal positive* between two basin-year entities of the same basin, and (3) a *spatial overlap positive* between entities with overlapping or nested basin boundaries. Other entity pairs without spatial overlap remain negative examples, but their repulsive strength is adjusted using basin-metadata similarity to be described below. Figure 2 summarizes these relations.

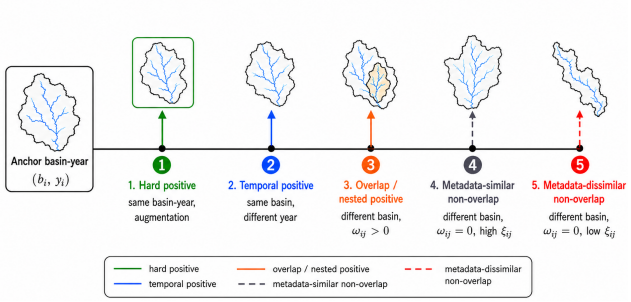


Figure 2. Basin relations used by the relation-aware contrastive objective.

For a mini-batch of B basin-year entities, two independently augmented views of each entity are generated and encoded into $N = 2B$ normalized embeddings $\{z_i\}_{i=1}^N$. Details of the augmentation procedure are provided in Appendix A.1. Each embedding is associated with a basin ID b_i, year y_i, and basin metadata $X_{b_i}^{\text{meta}}$. The generated embeddings form an anchor set $\mathcal{A} = \{1, \ldots, N\}$ for computing contrastive loss.

For each anchor $i \in \mathcal{A}$, we define the following relations:

$$\begin{aligned} \mathcal{P}_i^{\text{hard}} &= \{j : b_i = b_j,\ y_i = y_j,\ i \neq j\}, \\ \mathcal{P}_i^{\text{temp}} &= \{j : b_i = b_j,\ y_i \neq y_j\}, \\ \mathcal{P}_i^{\text{ov}} &= \{j : b_i \neq b_j,\ \omega_{ij} > 0\}, \\ \mathcal{N}_i^{\text{neg}} &= \{j : b_i \neq b_j,\ \omega_{ij} = 0\}. \end{aligned} \tag{7}$$

Here, $\mathcal{P}_i^{\text{hard}}$ contains the other augmented view of the same basin-year, $\mathcal{P}_i^{\text{temp}}$ contains other years of the same basin, $\mathcal{P}_i^{\text{ov}}$ contains spatially overlapping or nested basins, while $\mathcal{N}_i^{\text{neg}}$ contains different-basin entities with no spatial overlap. We quantify the spatial overlap between embeddings i and j using the Szymkiewicz–Simpson coefficient [32]:

$$\omega_{ij} = |G_i \cap G_j| / \min(|G_i|, |G_j|),$$

where G_i and G_j denote their corresponding basin polygons and $|\cdot|$ denotes polygon area. This coefficient reaches one when the smaller basin is fully nested within the larger, making it suitable for both overlapping and nested relations.

Based on the relation sets defined above, our relation-aware contrastive loss for anchor i is given by:

$$\mathcal{L}_i^{\text{con}} = -\sum_{j \neq i} q_{ij} \log \frac{d_{ij}\exp(s_{ij}/\tau)}{\sum_{k \neq i} d_{ik}\exp(s_{ik}/\tau)}, \tag{8}$$

where $s_{ij} = z_i^\top z_j$ is the cosine similarity between two normalized embeddings, q_{ij} measures the attraction (positive) strength between embeddings, d_{ij} measures the repulsive (negative) strength, while τ is a hyperparameter.

The attraction (positive) strength between embeddings i and j is computed as follows:

$$q_{ij} = \frac{w_{ij}^+}{\sum_{k \neq i} w_{ik}^+}, \qquad j \neq i. \tag{9}$$

where

$$w_{ij}^{+} = \mathbf{1}_{j\in\mathcal{P}_i^{\text{hard}}} + \lambda_t \kappa_{ij} \mathbf{1}_{j\in\mathcal{P}_i^{\text{temp}}} + \lambda_o \omega_{ij} \kappa_{ij} \mathbf{1}_{j\in\mathcal{P}_i^{\text{ov}}}, \quad (10)$$

Here, λ_t and λ_o control the contributions due to temporal and spatial overlap relations. The temporal factor $\kappa_{ij} = \exp(-|y_i - y_j|/\tau_y)$ decreases as the separation between years increases, with τ_y controlling the decay rate. This factor is applied to both temporal and spatial overlap relations to ensure that basin-year entities observed far apart in time have lower attraction strengths. In this formulation, q_{ij} is nonzero only for embedding pairs with positive relations. Since every basin-year entity has two augmented views, each anchor must have at least one hard positive.

To avoid excessive repulsion between physiographically similar basins, Basin2Vec assigns each pair the weight

$$d_{ij} = \begin{cases} 1, & w_{ij}^{+} > 0, \\ 1 - \lambda_m \xi_{ij}, & w_{ij}^{+} = 0, \end{cases} \quad (11)$$

where $\lambda_m \in [0, 1)$ controls the strength of metadata-based downweighting and $\xi_{ij} \in [0, 1]$ denotes the metadata similarity between the corresponding basins. The computation of ξ_{ij} is described in Appendix A.1. Positive pairs retain unit weight because their attraction is already controlled by w_{ij}^{+} and q_{ij}. For non-positive pairs, greater metadata similarity produces a smaller d_{ij} and therefore weaker repulsion. Metadata similarity is not treated as a positive relation since physiographic similarity does not necessarily imply similar hydrological behavior. The contrastive loss given in Equation (8) pulls each anchor toward its hard, temporal, and overlap positives while reducing unnecessary repulsion from metadata-similar non-overlapping basins.

The contrastive denominator provides relative negative pressure but does not directly bound the similarity of individual non-overlapping pairs. Basin2Vec therefore applies the following negative-margin loss

$$\mathcal{L}_{\text{neg}} = \frac{1}{|\mathcal{N}_{\text{neg}}|} \sum_{(i,j)\in\mathcal{N}_{\text{neg}}} \left[\max(0, s_{ij} - \mu_{ij})\right]^2, \quad (12)$$

where $\mathcal{N}_{\text{neg}} = \{(i, j) : i \in \mathcal{A},\ j \in \mathcal{N}_i^{\text{neg}}\}$ contains all non-overlapping negative pairs, and $\mu_{ij} = \mu_{\min} + (\mu_{\max} - \mu_{\min})\xi_{ij}$ is the metadata-dependent margin. Metadata-similar negative pairs receive a larger margin and may remain closer in the embedding space, whereas metadata-dissimilar pairs receive a stricter margin.

Altogether, the overall pre-training objective is

$$\mathcal{L} = \frac{1}{N} \sum_{i\in\mathcal{A}} \mathcal{L}_i^{\text{con}} + \lambda_n \mathcal{L}_{\text{neg}} + \lambda_a \mathcal{L}_{\text{area}}, \quad (13)$$

where the coefficients λ_n and λ_a control the contributions of the negative-margin and area-preservation terms, respectively. Note that $\mathcal{L}_i^{\text{con}}$ organizes the embedding space using known basin relations, $\mathcal{L}_{\text{neg}}$ limits excessive similarity between non-overlapping basins, while $\mathcal{L}_{\text{area}}$ encourages the encoder to retain information about physical basin scale.

3.4. Training and Embedding Generation

A basin-aware sampler selects the basins for each mini-batch and draws K_{yr} years per basin, to ensure the same number of temporal positives for each basin within a mini-batch. After training, the unaugmented input for each original basin-year entity is encoded to obtain its final embedding $z_{b,y}$.

Finally, to obtain a fixed descriptor for each basin, we take the average annual embeddings for each basin and ℓ_2-normalize the result:

$$\bar{z}_b = \text{Normalize}\left(\frac{1}{|\mathcal{Y}_b|} \sum_{y\in\mathcal{Y}_b} z_{b,y}\right), \quad (14)$$

where $\mathcal{Y}_b$ denotes the years available for basin b. This facilitates a fair downstream comparison since some EO baselines have different temporal coverages, i.e., do not provide embeddings for a common set of years as Basin2Vec.

4. Experimental Setup

We evaluate Basin2Vec through three research questions (RQs) covering intrinsic embedding structure, temporal generalization, and zero-shot spatial transfer.

4.1. RQ1: Intrinsic Embedding Structure

RQ1 tests whether the learned representations (i) remain diverse rather than collapsing into a low-dimensional subspace and (ii) preserve metadata-defined scale and shape relationships. For each method, we average the available embeddings per basin, apply ℓ_2 normalization, and evaluate the same 9,067 basins.

Representation diversity. We measure representation diversity using entropy-based effective rank [29]. Given the eigenvalues $\{\lambda_i\}_{i=1}^{d}$ of the covariance matrix of the embeddings, let $p_i = \lambda_i / \sum_{j=1}^{d} \lambda_j$. The effective rank of the learned embeddings is

$$r_{\text{eff}} = \exp\left(-\sum_{i=1}^{d} p_i \log p_i\right). \quad (15)$$

Since the embedding dimensions may differ for different methods, we report the normalized effective rank $\widetilde{r}_{\text{eff}} = r_{\text{eff}} / \min(d, N - 1)$, where d and N denote the embedding dimensions and number of basins, respectively. Higher values indicate that the embeddings are more diverse as their variance spans a larger fraction of the available dimensions.

Metadata consistency. The metadata incorporated into Basin2Vec include descriptors of basin scale and geometry (Appendix A.2), which influence basin hydrologic behavior. These descriptors are integrated through metadata fusion, while pairwise metadata similarity is used in

the relation-aware objective. We therefore test whether metadata-similar basins are closer in the learned embedding space than metadata-dissimilar basins.

We use standardized scale-and-shape metadata (Appendix A.2), with no geographic-coordinate inputs, and exclude overlapping and nested pairs. *Metadata-similar pairs* are defined using each basin's nearest neighbors in Euclidean metadata space. *Metadata-dissimilar pairs* are selected as the most distant pairs within a randomly sampled pool of non-overlapping candidates. The same pair sets are evaluated for every method. We compare the ℓ_2-distance distributions of normalized basin embeddings and summarize their separation as $\Delta_{\text{meta}} = \mu_{\text{dissimilar}} - \mu_{\text{similar}}$, where the two terms denote the respective mean pairwise distances. A larger positive Δ_{meta} indicates clearer preservation of metadata-defined basin relationships.

RQ2–RQ3: Downstream Streamflow Generalization

Task description. RQ2 and RQ3 evaluate basin representations as static conditioning vectors for daily streamflow prediction. For basin b and target day t, the downstream model predicts streamflow $Q_{b,t}$ from an input sequence of meteorological forcings $\mathbf{x}^{\text{met}}_{b,t}$ and a fixed basin descriptor c_b:

$$\widehat{Q}_{b,t} = F_\eta\left([\mathbf{x}^{\text{met}}_{b,t-L+1}, \ldots, \mathbf{x}^{\text{met}}_{b,t}], c_b\right). \tag{16}$$

where $L \in \{120, 365\}$ is the forcing-window length. Meteorological forcings are available through day t, but no observed streamflow is provided as input.

Data description. Experiments use CAMELS-US basins with Daymet meteorological forcings and USGS daily streamflow [1, 26]. The dynamic inputs are precipitation, shortwave radiation, maximum and minimum temperature, and vapor pressure. Discharge is converted to basin-mean runoff (mm/day); the 27 CAMELS attributes include drainage area (`area_gages2`). The main benchmark compares basin descriptors obtained from Basin2Vec, AlphaEarth, TESSERA, SatCLIP, and CAMELS attributes.

Because the learned representations differ in temporal coverage and spatial sampling, we construct one fixed descriptor per basin using all samples available from each source. Basin2Vec uses the ℓ_2-normalized, year-averaged representation $\bar{z}_b$ from Eq. (14). For the EO baselines, embeddings are averaged over the available years and sampled locations within each basin and then ℓ_2-normalized. This avoids selecting an arbitrary source-dependent year and provides a time-invariant descriptor comparable to the CAMELS attributes. All basin descriptors remain frozen during downstream training.

Evaluation protocol. For RQ2, *temporal generalization* is evaluated on the same 671 basins across disjoint periods. Data from 1990–2009 are used for model development, including early stopping, while 2010–2014 is held out for evaluation [10, 12].

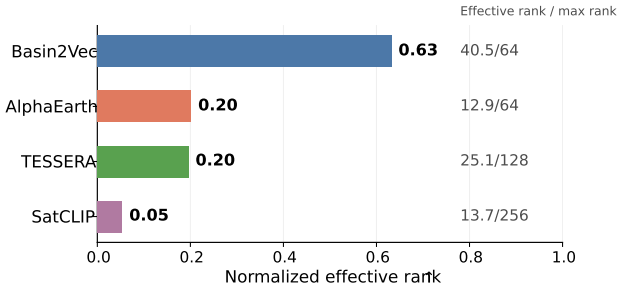


Figure 3. **Representation diversity.** Normalized effective rank across 9,067 basins. Higher values indicate greater representation diversity.

For RQ3, *zero-shot spatial transfer* is evaluated using 6-fold ungauged-basin cross-validation. In each fold, held-out basins are excluded from supervised training and early stopping. The remaining basins use 1990–2004 for training and 2005–2014 for validation, while the held-out basins are evaluated over 1990–2014 [3, 18, 31].

The downstream predictor F_η is instantiated as LSTM [15], TFT [20], TimeXer [37], Timer-XL [23], or iTransformer [22]. Within each architecture, the dynamic inputs, conditioning mechanism, and training protocol are held fixed across representation sources; only the descriptor input width changes to match the source dimension. All inputs, targets, and conditioning vectors are standardized using training data only.

We additionally compare Basin2Vec with KG-SSL [13], a hydrology-specific self-supervised method, using its unsupervised reconstruction-and-contrastive setting without static-attribute supervision. KG-SSL is trained and its descriptors are constructed from 1990–2009 meteorology and observed streamflow, then frozen for 2010–2014 evaluation. The comparison uses the matched downstream LSTM with $L = 365$ and lead-zero prediction. KG-SSL is excluded from RQ3 because its descriptor construction requires streamflow from the held-out basin.

Models are optimized using MSE and AdamW with validation-based early stopping and gradient clipping. Each configuration is evaluated using three seeds (42, 123, and 456). For each seed, NSE [25] and KGE [14] are computed per basin and summarized by their median across basins. We report the mean and standard deviation of these medians over the three seeds; higher values indicate better performance, with 1 representing perfect agreement.

Complete hyperparameter and implementation details are provided in the appendix.

5. Experimental Results

5.1. Intrinsic Embedding Structure *(RQ1)*

Basin2Vec achieves the highest normalized effective rank (Fig. 3), indicating that its variance is distributed across a larger fraction of the available dimensions rather than con-

Table 2. Temporal generalization results. Values are median basin-level NSE and KGE. **Bold** and underlining indicate the best and second-best conditioning sources, respectively.

Model	Median NSE ↑					Median KGE ↑				
	Basin2Vec	AlphaEarth	CAMELS Attr.	TESSERA	SatCLIP	Basin2Vec	AlphaEarth	CAMELS Attr.	TESSERA	SatCLIP
$L = 365$										
LSTM	**0.782±0.009**	0.641±0.017	0.747±0.010	0.662±0.025	0.719±0.018	**0.786±0.011**	0.572±0.013	0.756±0.013	0.626±0.022	0.688±0.031
TFT	**0.753±0.011**	0.741±0.026	0.717±0.022	0.570±0.025	0.638±0.026	**0.768±0.019**	0.722±0.029	0.714±0.030	0.514±0.033	0.566±0.026
TimeXer	**0.651±0.018**	0.630±0.012	0.502±0.019	0.573±0.033	0.603±0.015	**0.691±0.009**	0.579±0.023	0.446±0.030	0.517±0.023	0.571±0.033
Timer-XL	**0.730±0.019**	0.674±0.028	0.591±0.026	0.627±0.019	0.402±0.021	**0.708±0.022**	0.585±0.010	0.536±0.029	0.635±0.021	0.346±0.016
iTrans.	**0.627±0.022**	0.559±0.020	0.555±0.022	0.576±0.015	0.464±0.017	**0.582±0.026**	0.566±0.024	0.533±0.020	0.579±0.028	0.450±0.022
$L = 120$										
LSTM	**0.754±0.014**	0.619±0.015	0.737±0.013	0.638±0.016	0.704±0.030	**0.775±0.009**	0.563±0.023	0.741±0.031	0.592±0.031	0.705±0.025
TFT	**0.737±0.019**	0.719±0.020	0.712±0.012	0.551±0.029	0.668±0.026	**0.744±0.020**	0.719±0.032	0.668±0.030	0.479±0.014	0.600±0.020
TimeXer	**0.677±0.011**	0.675±0.010	0.645±0.026	0.592±0.014	0.530±0.028	**0.679±0.025**	0.625±0.016	0.675±0.011	0.563±0.028	0.492±0.010
Timer-XL	**0.705±0.018**	0.650±0.026	0.529±0.021	0.602±0.019	0.566±0.016	**0.693±0.014**	0.580±0.016	0.440±0.029	0.540±0.027	0.533±0.025
iTrans.	**0.594±0.019**	0.551±0.015	0.581±0.028	0.584±0.017	0.398±0.025	**0.580±0.009**	0.545±0.010	0.554±0.012	0.535±0.031	0.349±0.024

Table 3. Zero-shot spatial-transfer results for $L = 365$. Each value is the median per-basin NSE or KGE on held-out basins. **Bold** and underlining indicate the best and second-best conditioning sources, respectively.

Model	Median NSE ↑					Median KGE ↑				
	Basin2Vec	AlphaEarth	CAMELS Attr.	TESSERA	SatCLIP	Basin2Vec	AlphaEarth	CAMELS Attr.	TESSERA	SatCLIP
LSTM	**0.460±0.050**	0.079±0.021	0.071±0.018	0.092±0.017	0.132±0.022	**0.408±0.001**	−0.084±0.025	−0.098±0.027	−0.049±0.020	−0.072±0.024
TFT	**0.524±0.019**	0.132±0.017	0.147±0.015	0.127±0.018	0.119±0.021	**0.427±0.022**	−0.077±0.028	−0.071±0.025	−0.055±0.019	−0.100±0.023
TimeXer	**0.491±0.016**	0.126±0.020	0.065±0.018	0.065±0.021	0.106±0.017	**0.431±0.018**	−0.050±0.022	−0.080±0.019	−0.055±0.020	−0.066±0.024
Timer-XL	**0.498±0.017**	0.136±0.024	0.092±0.019	0.022±0.016	0.140±0.021	**0.436±0.020**	−0.038±0.021	−0.060±0.023	−0.114±0.029	−0.023±0.018
iTrans.	**0.465±0.021**	0.151±0.018	0.007±0.014	0.085±0.017	0.058±0.020	**0.384±0.024**	−0.039±0.020	−0.093±0.026	−0.098±0.024	−0.114±0.028

Table 5. Comparison with KG-SSL under the matched LSTM setting with $L = 365$.

Method	Median NSE ↑	Median KGE ↑
KG-SSL	0.620±0.018	0.593±0.015
Basin2Vec	**0.782±0.009**	**0.786±0.011**

centrated in a small subspace. All methods place metadata-dissimilar pairs farther apart on average than metadata-similar pairs, but Basin2Vec exhibits the largest separation Δ_{meta}, while the baseline distributions overlap more substantially (Fig. 4). These results show that Basin2Vec produces a diverse, non-collapsed representation that more clearly preserves basin scale and geometry relationships.

5.2. Temporal Generalization *(RQ2)*

Table 2 summarizes the temporal generalization performance of the embedding methods across five downstream architectures. Basin2Vec ranks first in both NSE and KGE for every architecture at forcing-window length $L \in \{120, 365\}$. Its consistent gains across recurrent, attention-based, and transformer-style predictors indicate that the benefit is not specific to one architecture or forcing-window length. The right-shifted KGE distributions in Figure 5 further show that the gains extend across basins.

Comparison with hydrology-specific SSL. With the same LSTM and $L = 365$, Basin2Vec increases median NSE from 0.620 to 0.782 and median KGE from 0.593 to

Table 4. Component ablations of Basin2Vec using LSTM with $L = 365$. **Bold** and underlining denote the best and second-best results.

Category	Variant	Ablation setting	Temporal		Spatial	
			NSE ↑	KGE ↑	NSE ↑	KGE ↑
Full model	**Basin2Vec**	All components	$\mathbf{0.782 \pm 0.009}$	$\mathbf{0.786 \pm 0.011}$	$\mathbf{0.460 \pm 0.050}$	$\mathbf{0.408 \pm 0.001}$
Scale-aware encoder	w/o monthly branch	$z^{\text{dyn}}_{b,y}$ removed	0.649 ± 0.016	0.621 ± 0.031	0.423 ± 0.021	0.388 ± 0.032
	w/o quasi-static branch	$z^{\text{sta}}_{b,y}$ removed	0.635 ± 0.015	0.623 ± 0.014	0.404 ± 0.013	0.397 ± 0.007
	w/o metadata conditioning	$\text{MLP}_{\text{meta}}()$ and $g_{b,y}$ removed	0.634 ± 0.022	0.615 ± 0.017	0.064 ± 0.044	-0.059 ± 0.044
	w/o gated metadata fusion	gated fusion → concat. + MLP	0.651 ± 0.032	0.620 ± 0.059	0.424 ± 0.029	0.399 ± 0.042
	w/o area regularization	$\lambda_a = 0$	0.601 ± 0.007	0.571 ± 0.001	0.412 ± 0.013	0.389 ± 0.014
Relation-aware objective	hard positive only	$w^+_{ij} = \mathbf{1}_{j \in \mathcal{P}^{\text{hard}}_i},\ d_{ij} = 1$	0.670 ± 0.009	0.647 ± 0.021	0.372 ± 0.010	0.332 ± 0.021
	w/o temporal positives	$\lambda_t = 0$	0.621 ± 0.024	0.628 ± 0.056	0.418 ± 0.030	0.398 ± 0.011
	w/o overlap/nesting positives	$\lambda_o = 0$	0.633 ± 0.012	0.589 ± 0.001	0.408 ± 0.017	0.389 ± 0.017
	w/o metadata-weighted denominator	$d_{ij} = 1$	0.638 ± 0.028	0.608 ± 0.035	0.384 ± 0.017	0.330 ± 0.008
	metadata as positive	$w^+_{ij} \mathrel{+}= \lambda_p \xi_{ij}$	0.600 ± 0.039	0.580 ± 0.040	0.403 ± 0.002	0.390 ± 0.004
	w/o negative-margin loss	$\lambda_n = 0$	0.643 ± 0.032	0.625 ± 0.048	0.409 ± 0.024	0.383 ± 0.029

0.786 over KG-SSL [13] (Table 5). Although KG-SSL uses streamflow, it learns from lumped one-year sequences over 671 basins, making its descriptors more dependent on the observed flow regime. Basin2Vec instead learns from distributed inputs across 9,067 basins and enforces consistency across years, producing more stable descriptors for temporal transfer.

5.3. Zero-Shot Spatial Transfer *(RQ3)*

Table 3 shows that Basin2Vec ranks first in NSE and KGE for every downstream architecture. Its median NSE exceeds the strongest alternative for each architecture by 0.314–0.377, and it is the only method with consistently positive median KGE, demonstrating substantially stronger transfer to unseen basins. We also examine the relative performance of the methods across different basin sizes. Figure 6 further shows that Basin2Vec maintains the strongest NSE and KGE distributions for small, medium, and large basins. Thus, the gains are not confined to a particular basin-size range.

5.4. Ablation Study

We remove or replace individual components of the scale-aware encoder and relation-aware objective and evaluate the resulting variants under temporal and spatial generalization. Table 4 shows that the complete Basin2Vec model achieves the best result for every metric. *Scale-aware encoder.* Removing either input branch degrades generalization, while removing metadata conditioning causes the largest spatial decline and negative median KGE. Gated metadata fusion outperforms concatenation, and area regularization provides particularly strong temporal gains. *Relation-aware objective.* The full objective outperforms every relation ablation. Hard positives alone rank second temporally but transfer poorly to unseen basins. Removing temporal or overlap/nesting positives, metadata-weighted negatives, or the negative-margin loss reduces performance. Treating metadata as a positive relation also performs worse, supporting its use to moderate negative repulsion rather than enforce attraction.

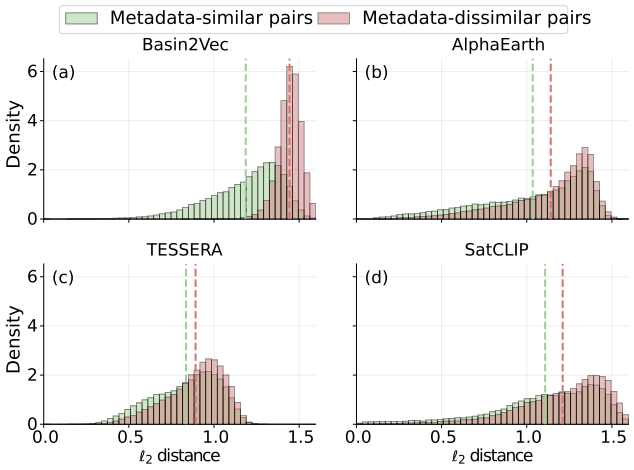


Figure 4. **Metadata consistency.** ℓ_2-distance distributions for metadata-similar and metadata-dissimilar non-overlapping pairs. Dashed lines denote means.

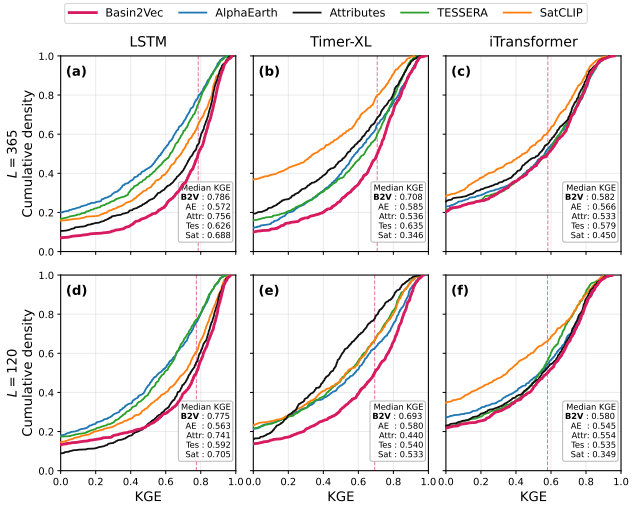


Figure 5. Per-basin KGE CDFs for temporal generalization using selected downstream architectures.

6. Conclusion

We presented Basin2Vec, a self-supervised framework for learning basin-centric representations from gridded hydroclimatic data without observed streamflow. Basin2Vec combines a scale-aware encoder for spatiotemporal inputs and metadata with a relation-aware contrastive objective that models hydrologic relationships across space and time. The resulting embeddings are diverse and hydrologically structured. Across architectures and forcing windows, Basin2Vec outperforms predefined attributes and general-purpose EO embeddings in temporal generalization and zero-shot transfer to unseen basins. Ablations validate its main design choices. These results show that explicitly modeling basin structure and relationships yields reusable representations for hydrologic prediction.

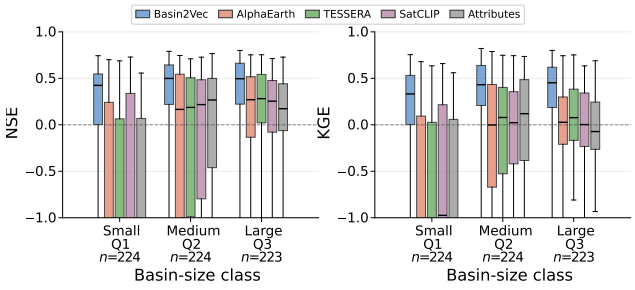


Figure 6. Spatial-split performance across basin-size classes.

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